



Nearby stars could host life

A planet hunting algorithm reveals that 9 percent of stars nearby could host planets. <https://digit.in/july19-42>



Uranium Ring System clicked

Thermal images of the Uranus Rings recorded temperatures of -321 degrees Fahrenheit. <https://digit.in/july19-43>

more armour. One of the institutes was the Carnegie Mellon University, which came up with an innovative approach of using shape shifting wheels.

For armoured vehicles, there is no single best answer to whether tracks are better or wheels are better. While tracks perform better on soft surfaces, wheels perform better on hard surfaces. A combat vehicle is expected to perform on a wide range of terrain. The National Robotics Engineering Center (NREC) at CMU developed tyres that could be reconfigured on the fly, and switch from a triangular track configuration to a wheel configuration on demand. The core principle at work here is the amount of surface area of the wheel that is in contact with the road at any point of time. More surface area means more traction, which is an advantage in off road situations. Dimi Apostolopoulos, the senior scientist who led the project explains, "This shape-changing locomotion technology could enable vehicles to tackle a wide array of terrains at surprising speeds. Based on the testing we've done so far, we would expect such a vehicle to do amazing things." In the track mode, the vehicle achieved a top speed of 48 kmph, and 80 kmph in the wheeled mode. The vehicles need to slow down for the switch in configurations, but do not need to stop moving. The tyres can go from track to wheel mode at 40 kmph, and from wheel mode to track mode at 19 kmph. The design is referred to as a reconfigurable wheel-track or RWT for short. Initially, the approach used a set of electric motors to change the shape of the wheel, but then realised that the same result could be achieved by extending a Y shaped support that changes the shape from wheel to track instantly and passively. At the same time, a brake is applied that causes the transmission to start moving a set of gears instead of the wheels. These gears drive the track. RWTs can switch from the wheeled mode to the track mode and back again within 2 seconds. The reconfigurable wheels can be mounted on a vehicle in various

combinations. This can be all four wheels, just the front two wheels, or just the back wheels. The technology has potential applications beyond defence, including in forestry, mining and construction.

TO INFINITY AND BEYOND

From the 1960s, the development of new types of wheels are an area of research for space agencies. This is because new types of wheels had to be developed for moving on the surface of the Moon, and work in conditions of near vacuum. The Russian Lunokhod program sent a number of robotic explorers to the surface of the Moon, and the Lunokhod 1 was the first mobile explorer on another world. It used a set of wire mesh wheels, each driven by an independent electric motor. NASA used nitrogen filled tyres



NASA's superelastic tyres

made by Goodyear for hand carts used to transport equipment. The moon buggies, technically called the Lunar Roving Vehicle used wire mesh wheels with stiff frames to prevent too much deformation. The Curiosity Rover on Mars used a stiff wheel design made out of aluminum, which turned out to be problematic. The wheels developed holes and tears because of the harsh terrain, and negatively affected the duration of the mission.

For future Mars Rovers, NASA engineers are considering revisiting the wire mesh wheels. This model, which is a more advanced version of the bicycle tyre, was developed in the mid 2000s by Vivake Asnani, a NASA engineer in collaboration with Goodyear. The initial design involved

hundreds of coiled steel fibres wound up into a mesh, which is flexible and yet resilient to changes in shape when the pressures are reduced. However, when tested on simulated Martian terrain, the steel tyres showed too much deformation. The breakthrough came when the same design was used with a different material - nickel titanium, which is a shape memory alloy. The tyres returned to their original shape after deformation. The properties allow the tyre to be pliant on harsh terrain, and support heavy loads as they conform to the terrain instead of sinking. The tyre also has the advantage of providing ample traction on soft terrain. These tyres can allow future robotic explorers to access areas previously out of reach on the Moon and Mars. The tyres are also expected to be used in future manned explorations, where the vehicles will be required to support more weight. This innovation is called the superelastic tyre.

While the gear made for Earth might not work very well in the extreme conditions on the Moon or Mars, the technology developed for space exploration can definitely be used on Earth. NASA is in the technology transfer process of licensing its superelastic tyres, so one day you might be able to purchase an ATV that has the same tyres as the future rovers on Mars.

Non-pneumatic tyres are used in construction equipment, bicycles and wheelchairs with varying degrees of success. They have not made their way into vehicles because of stability and safety issues, especially at high speeds. On 19 June 2019, Michelin announced plans to bring airless tyres to cars by 2024, in partnership with General Motors. These tyres do not get punctured, have no air pressure to lose, and require very little maintenance. The development could also reduce the vast number of waste tyres ending up in junkyards around the world. The airless wheel technology is called Unique Punctureproof Tire System or Uptis by Michelin, and is an evolution of their Tweel concept. 